REPS Digital Twin – ME-2

Physical Baseline Data Collection Specification

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1 Purpose

The purpose of this data collection is to construct a structured physical-response database for training and validating the behavioral digital twin of the laboratory bridge.

The database will characterize how the healthy bridge responds as the following physical variables are changed independently and in combination:

- Applied load magnitude, L
- Applied load position, x_L
- Vibration excitation frequency, f_v
- Vibration excitation amplitude / motor command, A_v
- Left, center, and right bridge temperatures, T_L, T_C, and T_R

The resulting digital twin will represent the expected healthy response as

$$\boxed{\hat{\mathbf{y}} = f(L, x_L, f_v, A_v, T_L, T_C, T_R)}$$

where the measured sensor response is initially

$$\mathbf{y} = [LC_L,\ LC_R,\ a_x,\ a_y,\ a_z].$$

Processed dynamic features such as acceleration RMS and dominant frequency can later be extracted from the raw accelerometer measurements.

The purpose of the twin is therefore not simply to reproduce sensor data, but to predict what the physical bridge *should* be doing under a known set of operating conditions. Future measured behavior can then be compared against this expected response.

2 Sensor Layout

The bridge is instrumented as shown in Table 1.

Table 1: Bridge Instrumentation

Sensor	**Location**	**Measurement**
Left Load Cell	Left side of bridge	Left-side structural load response
Right Load Cell	Right side of bridge	Right-side structural load response
Accelerometer	Center of bridge	Acceleration in x, y, and z directions
Vibration Motor	Center of bridge	Controlled dynamic excitation
Temperature Sensor 1	Left load-cell location	Local temperature near left load cell
Temperature Sensor 2	Center of bridge	Center structural temperature
Temperature Sensor 3	Right load-cell location	Local temperature near right load cell

All three temperature measurements must remain separate. They should not be averaged during raw data collection because a temperature gradient across the bridge may affect the two load cells differently.

3 Fixed Experimental Levels

The experimental levels used during the Fall 2026 baseline collection are defined in Table 2.

Table 2: Baseline Experimental Levels

Variable	**Defined Levels**
Load Magnitude	$L00 = 0$; $L01 = L_{\text{low}}$; $L02 = L_{\text{medium}}$; $L03 = L_{\text{high}}$
Load Position	$PL = 0.25L_b$; $PC = 0.50L_b$; $PR = 0.75L_b$
Vibration Amplitude	$V00 = 0$; $V01 = 64$; $V02 = 128$; $V03 = 192$ PWM command
Temperature	$TEMP01$ = room/baseline; $TEMP02$ = warm; $TEMP03$ = hot
Calibration	Nominal sensor calibration unless explicitly testing calibration sensitivity

The three nonzero motor commands correspond approximately to 25%, 50%, and 75% of the available 0–255 command range:

$$\frac{64}{255} \approx 25\%, \qquad \frac{128}{255} \approx 50\%, \qquad \frac{192}{255} \approx 75\%.$$

The actual numerical load values used for $L01$, $L02$, and $L03$ must be recorded in the master test index.

4 Separating Vibration Amplitude and Frequency

Vibration amplitude and vibration frequency must be treated as separate experimental variables.

During the primary vibration-amplitude baseline, the excitation frequency is held constant while the motor command is changed:

$$\boxed{f_v = 10 \text{ Hz}}$$

with

$$A_v \in \{64,\ 128,\ 192\}.$$

This isolates the effect of vibration amplitude on the measured bridge response.

A separate frequency sweep is then performed at a fixed motor command:

$$\boxed{A_v = 128}$$

with commanded frequencies

$$\boxed{f_v \in \{5,\ 10,\ 15 \text{ Hz}\}.}$$

The purpose of separating these tests is to avoid changing amplitude and frequency simultaneously. If both variables are changed in the same test series, their individual effects on bridge response cannot be clearly distinguished.

The accelerometer FFT should also be used later to calculate the *measured* dominant structural frequency, f_{dom}. The commanded motor frequency and measured dominant frequency should remain separate variables in the processed dataset.

5 Data File Naming Standard

Each experimental run must be stored as an individual CSV file using

`ME2_T####_L##_P##_V##_TEMP##.csv`

where:

- T#### = sequential test number
- L00--L03 = load magnitude
- P00 = no applicable load position
- PL, PC, PR = left, center, right
- V00--V03 = vibration amplitude level
- TEMP01--TEMP03 = temperature condition

For example:

ME2_T0001_L00_P00_V00_TEMP01.csv

Repeated tests receive a new test number while all physical-condition codes remain unchanged. For example, three repetitions of the same condition are:

ME2_T0001_L00_P00_V00_TEMP01.csv
ME2_T0002_L00_P00_V00_TEMP01.csv
ME2_T0003_L00_P00_V00_TEMP01.csv

Physical-event datasets may use an additional suffix:

_WIND
_IMPACT

6 Raw CSV Format

Every raw file must contain the same channels in the same order.

Table 3: Raw CSV Column Standard

Column	Field	Description
1	time_s	Time from beginning of test [s]
2	load_left	Left load-cell measurement
3	load_right	Right load-cell measurement
4	accel_x_g	X-axis acceleration [g]
5	accel_y_g	Y-axis acceleration [g]
6	accel_z_g	Z-axis acceleration [g]
7	temp_left_C	Left temperature [°C]
8	temp_center_C	Center temperature [°C]
9	temp_right_C	Right temperature [°C]
10	vibration_input	Motor command, 0–255

Column headers must always be included. Raw files should not contain titles, comments, blank rows, merged cells, or manually formatted sections.

Raw data should not be manually altered after collection.

7 Master Test Index

A separate file should be maintained as

`ME2_test_index.csv`.

Each row corresponds to one experimental run.

Table 4: Master Test Index

Field	Description
`test_id`	Unique sequential test identifier
`filename`	Corresponding raw CSV filename
`duration_s`	Recorded test duration
`load_value`	Actual applied load
`load_position`	Measured physical load position
`vibration_level`	V00, V01, V02, or V03
`vibration_command`	Actual 0–255 motor command
`commanded_frequency_Hz`	Commanded vibration frequency
`temperature_condition`	TEMP01, TEMP02, or TEMP03
`calibration_state`	Nominal or defined altered calibration state
`scenario`	Normal, wind_gust, impact, or other defined physical event
`notes`	Anything unusual observed during the experiment

8 Test Duration Standard

Table 5: Standard Test Durations

Test Type	**Duration**	**Purpose**
No-load baseline	30 s	Unloaded sensor baseline
Vibration amplitude test	30 s	Response versus vibration amplitude
Frequency sweep point	30 s	Response versus excitation frequency
Static load test	30 s	Steady load distribution
Combined load + vibration	30 s	Simultaneous static/dynamic response
Calibration / zero check	20 s	Sensor zero and baseline stability
Temperature steady-state	60 s	Response at stabilized thermal condition
Temperature transient	2–5 min	Thermal drift during heating/cooling
Wind/gust simulation	30 s	Physical disturbance and recovery
Impact simulation	15 s	Impact transient and ring-down

At an approximate sampling rate of 100 Hz, a 30-second run contains

$$N = F_s t = (100)(30) = 3000$$

samples per continuously sampled sensor channel.

9 Standard 30-Second Window

Normal 30-second experiments should use

0–5 s : settling

5–25 s : steady-state analysis

25–30 s : end buffer.

The requested physical condition should already be established before the 5–25 second analysis interval.

10 Stage 1 – No-Load Vibration-Amplitude Baseline

The first experiment isolates vibration amplitude with no applied load.

The vibration frequency is fixed at

$$f_v = 10 \text{ Hz}.$$

Four amplitude conditions are tested:

Table 6: No-Load Vibration-Amplitude Baseline

Code	Motor Command	Frequency	Repeats
V00	0	Off	3
V01	64	10 Hz	3
V02	128	10 Hz	3
V03	192	10 Hz	3

Each run lasts 30 seconds, producing 12 files and

$$12(30) = 360 \text{ s} = 6 \text{ min}$$

of recorded data.

The corresponding filenames begin:

```
ME2_T0001_L00_P00_V00_TEMP01.csv
ME2_T0002_L00_P00_V00_TEMP01.csv
ME2_T0003_L00_P00_V00_TEMP01.csv
                ⋮
ME2_T0012_L00_P00_V03_TEMP01.csv
```

11 Stage 2 – Vibration Frequency Sweep

A separate frequency sweep is performed so that excitation-frequency effects are not mixed with vibration-amplitude effects.

For this experiment:

$$L = 0, \qquad A_v = 128.$$

The commanded frequencies are

$$f_v = 5,\ 10,\ 15 \text{ Hz}.$$

Table 7: No-Load Frequency Sweep

Frequency	**Motor Command**	**Repeats**
5 Hz	128	3
10 Hz	128	3
15 Hz	128	3

Each frequency is recorded for 30 seconds and repeated three times.

This produces

$$3(3)(30) = 270 \text{ s} = 4.5 \text{ min}$$

of frequency-sweep data.

The measured accelerometer response should later be transformed using an FFT to determine the actual dominant response frequency:

$$f_{\text{dom}} = \arg\max_{f} |X(f)| .$$

This allows commanded excitation frequency and measured structural response frequency to be compared directly.

12 Stage 3 – Static Load Matrix

Static testing is performed with the vibration motor off:

$$A_v = 0.$$

Three nonzero load magnitudes are used:

$$L01 = L_{\text{low}}, \qquad L02 = L_{\text{medium}}, \qquad L03 = L_{\text{high}}.$$

Each load is tested at three marked positions:

$$PL = 0.25L_b, \qquad PC = 0.50L_b, \qquad PR = 0.75L_b.$$

The full static matrix is therefore:

Table 8: Static Load Matrix

Load	Left	Center	Right
L01 – Low	3 runs	3 runs	3 runs
L02 – Medium	3 runs	3 runs	3 runs
L03 – High	3 runs	3 runs	3 runs

This gives

$$3 \text{ loads} \times 3 \text{ positions} \times 3 \text{ repeats} = 27 \text{ runs}.$$

At 30 seconds per run,

$$27(30) = 810 \text{ s} = 13.5 \text{ min}.$$

These experiments characterize both load magnitude sensitivity and load distribution between the two load cells.

13 Stage 4 – Combined Load and Vibration

The combined-response experiments characterize how dynamic excitation modifies the measured response while the bridge is carrying load.

The three vibration amplitudes are

$$A_v = 64,\ 128,\ 192$$

while excitation frequency is maintained at

$$f_v = 10 \text{ Hz}.$$

The preferred full experimental matrix is

$$3 \text{ loads} \times 3 \text{ positions} \times 3 \text{ vibration amplitudes}.$$

Each condition should be repeated three times.

This produces

$$3 \times 3 \times 3 \times 3 = 81 \text{ runs}.$$

At 30 seconds per run,

$$81(30) = 2430 \text{ s} = 40.5 \text{ min}.$$

If laboratory time becomes limited, the medium load condition should be prioritized across all three positions and all three vibration amplitudes before reducing the number of repeated baseline measurements.

14 Stage 5 – Temperature Variation

Three thermal conditions are defined:

- `TEMP01`: room/baseline condition
- `TEMP02`: warm condition
- `TEMP03`: highest safely achievable hot condition

The actual measured temperatures, rather than the categorical labels alone, will be used for digital twin training.

For every thermal test, all three temperature channels must be recorded:

$$T_L, \qquad T_C, \qquad T_R.$$

After reaching an approximately steady thermal condition, record for

$$\boxed{60 \text{ s}}.$$

At minimum, each thermal condition should include:

- No load, vibration off
- Medium center load, vibration off
- Medium left load, vibration off
- Medium right load, vibration off
- Medium center load + $A_v = 128$ at 10 Hz

A continuous heating or cooling transient should additionally be recorded for

$$\boxed{2\text{–}5 \text{ min}}.$$

This transient dataset is particularly useful for identifying temperature drift in the load-cell and accelerometer measurements.

15 Stage 6 – Physical Event Data

Physically abnormal but non-cyber events should be recorded after the normal baseline database has been established.

These data provide examples of legitimate physical anomalies so that later AI models do not learn that every unusual sensor response represents a cyberattack.

Wind/Gust

Use a 30-second recording:

0–5 s : normal baseline

5–20 s : controlled wind/gust disturbance

20–30 s : recovery.

The scenario label should be

`scenario = wind_gust`.

A corresponding filename may be

`ME2_TXXXX_L00_P00_V00_TEMP01_WIND.csv`.

Vehicle Impact / Crash

Use a 15-second recording:

0–4 s : pre-impact baseline

$t \approx 4$ s : controlled impact

4–15 s : transient decay and recovery.

The scenario label should be

`scenario = impact`.

A corresponding filename may be

`ME2_TXXXX_L02_PC_V00_TEMP01_IMPACT.csv`.

All impact testing must remain controlled and below a level that could damage the bridge or instrumentation.

These tests establish the distinction

$$\boxed{\text{Physically Abnormal} \neq \text{Cyberattack}}.$$

16 Load Position Standard

The load locations should be physically marked on the bridge rather than estimated visually during each run.

Using the usable bridge span L_b:

$$x_L = 0.25L_b, \qquad x_C = 0.50L_b, \qquad x_R = 0.75L_b.$$

The actual measured distance corresponding to each location should also be recorded.

Once marked, the same physical locations should be used throughout all experiments.

17 Repeatability Standard

Each important baseline condition should be collected as three separate runs.

Three 30-second files are preferred over one continuous 90-second file because this allows both within-run and between-run repeatability to be evaluated.

For a measured quantity y, the mean is

$$\mu = \frac{1}{N}\sum_{i=1}^{N} y_i$$

and standard deviation is

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \mu)^2}.$$

These statistics can later be used to establish expected operating bounds for the digital twin.

18 Standard Test Procedure

Every normal experimental run should follow the same general procedure:

1. Verify that all sensors are connected and producing reasonable data.
2. Verify the zero/calibration condition of both load cells.
3. Assign the next sequential test number.
4. Record the intended test condition in the master index.
5. Place the load at the appropriate marked location.
6. Set the vibration frequency and motor command.
7. Allow approximately 5 seconds for the physical system to settle.
8. Begin synchronized recording of all channels.
9. Maintain the exact test condition for the required duration.
10. Do not move the load or change vibration settings during a steady-state run.
11. Stop recording before changing the physical condition.
12. Save the raw CSV using the standardized filename.
13. Check the saved file for missing or obviously invalid measurements.
14. Proceed to the next test.

The sampling rate and sensor configuration should remain unchanged throughout the experimental campaign whenever possible.

19 Digital Twin Training Format

After collection, each experimental run can be converted into input and output features.

The primary digital twin input vector is

$$\boxed{\mathbf{x} = [L,\, x_L,\, f_v,\, A_v,\, T_L,\, T_C,\, T_R]}$$

and a useful processed output vector is

$$\boxed{\mathbf{y} = [LC_L,\, LC_R,\, a_{\mathrm{RMS}},\, f_{\mathrm{dom}}].}$$

Acceleration magnitude may first be calculated as

$$a(t) = \sqrt{a_x^2(t) + a_y^2(t) + a_z^2(t)}.$$

The RMS dynamic response is then

$$a_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} a_i^2}.$$

The digital twin learns the expected healthy mapping

$$\hat{\mathbf{y}} = f(\mathbf{x}).$$

During later operation, the measured response is compared with the twin prediction:

$$\boxed{\mathbf{r} = \mathbf{y}_{\mathrm{measured}} - \hat{\mathbf{y}}}.$$

These residuals quantify how far the real bridge response deviates from the expected healthy physical response.

The 5–25 second interval of each normal 30-second test should be used as the default steady-state feature-extraction window.

20 CpE Data Coordination

The ME and CpE teams should use a shared data structure that preserves the difference between physical conditions, physical measurements, digital-twin outputs, and cyberattack information.

The following categories should remain independently identifiable:

- Physical operating conditions
- Raw sensor measurements
- Extracted physical features
- Digital twin expected values
- Digital twin residuals
- Physical anomaly/event labels
- Cyberattack labels

The ME dataset should remain the physical ground truth. Cyberattack manipulations and labels can then be added by the CpE teams without modifying or destroying the original clean measurements.

21 Minimum Data Collection Rules

For every experiment:

1. Use one CSV per experimental run.
2. Use the exact same sensor columns and column order.
3. Assign a unique sequential test number.
4. Use the standardized filename.
5. Record the experiment in `ME2_test_index.csv`.
6. Record actual load values, not only low/medium/high labels.
7. Record actual vibration command values from 0–255.
8. Record commanded vibration frequency.
9. Keep amplitude fixed during frequency-sweep tests.
10. Keep frequency fixed during amplitude-sweep tests.
11. Use the same physically marked load positions.
12. Record all three temperature channels during every test.
13. Do not change operating conditions during a steady-state run.
14. Keep the sampling rate consistent.
15. Repeat primary baseline conditions three times.
16. Never overwrite or manually modify the original raw data.

Following these standards will produce a multidimensional healthy-response database that can be used to train the bridge digital twin, quantify normal physical variability, distinguish legitimate physical disturbances from cyber-induced behavior, and provide physically meaningful features to the CpE AI-monitoring pipeline.